HARIKRISHNAN S

B.Tech Undergraduate — Computer Science & Engineering

+91 7012249015 | sreehari007atl@gmail.com | linkedin.com/in/harikrishnanS05 | github.com/Harikrishnan0000

Chirayinkeezhu, Trivandrum, Kerala — 695304

EDUCATION

Mar Baselios College of Engineering & Technology (Autonomous) *2023 – 2027 (4th Year)* ***CGPA : 9.36***

B.Tech in Computer Science and Engineering | Trivandrum, Kerala

Jyothis Central School *2021 – 2023* ***Percentage: 91%***

Higher Secondary (11th & 12th) | Trivandrum, Kerala

Sree Chithra Public School *2011 – 2021*

Secondary Education (1st – 10th) | Trivandrum, Kerala

TECHNICAL SKILLS

Programming Languages: Python, C, Java

Web Development: MERN Stack (MongoDB, Express.js, React, Node.js), RESTful APIs, SQL

AR & Mobile Development: React Native, Unity Engine, ARCore, Flutter

Soft Skills: Problem Solving, Critical Thinking, Quick Learner, Adaptability

INTERNSHIP EXPERIENCE

Full Stack Development Internship — ICT Academy of Kerala *Dec 2024 – Jan 2025*

- Completed a MERN stack internship gaining hands-on experience in building and deploying dynamic web applications.
- Developed RESTful APIs and managed databases with a focus on clean code, security, and performance.

PROJECTS

SmartNav360 – AI-Assisted 360° Virtual Navigation Framework (Ongoing)

- Developing a browser-based framework that lets non-specialist staff build and publish navigable 360° virtual tours of campuses and buildings using only a smartphone and a desktop browser.
- Built a QR-linked capture session that pairs a phone camera with the desktop workspace, so overlapping photos upload and sync in real time without a native app.
- Implemented in-browser panorama stitching with OpenCV.js (WebAssembly), using ORB feature detection, Lowe's ratio-test matching, RANSAC homography estimation and perspective warping, so no server-side computer vision is needed.
- Integrated Marzipano for interactive 360° rendering, where author-placed hotspots act as clickable links and as weighted edges of a scene graph (scenes as nodes).
- Implemented Dijkstra's shortest-path routing over the scene graph, with the viewer highlighting the next hotspot at each step for turn-by-turn guidance.
- Added a generative-AI floor-plan renderer that converts 2D plans into a top-down minimap, plus optional object-detection and LLM-based assistant endpoints for scene labels and project Q&A.

- Built a modular full-stack back end with Node.js, Express and MongoDB, using multi-document transactions and a durable deferred-deletion queue to keep database records and uploaded files consistent.

Tech Stack: React.js (Vite), Ant Design, Node.js, Express.js, MongoDB (Mongoose), Marzipano, OpenCV.js/WebAssembly, JavaScript, HTML/CSS, Dijkstra's algorithm, LLM and object-detection APIs, Git..

Litter Detection System (AI-Powered Garbage Monitoring)

- Developed an AI-powered waste detection solution that analyzes CCTV footage using YOLOv11 for real-time object detection and BoT-SORT for multi-object tracking, so that people and objects are followed consistently across video frames.
- Integrated a Ollama with moondream VLM model for semantic verification and automated description generation of detected littering incidents, helping to filter out false detections and producing clear, human-readable incident summaries.
- System generates real-time alerts and detailed reports for monitoring and management of improper garbage disposal, helping authorities identify incidents quickly and take timely action.

- Applied multi-object tracking to assign persistent IDs to detected objects, enabling the system to link a discarded item to the moment and person involved in the incident rather than treating each frame in isolation.
- Structured the pipeline as detection, tracking, VLM verification, and alert generation stages, making each component independently testable and easy to upgrade.

AR Navigation System (Phase I)

- Built an AR-based navigation application using Unity Engine, and ARCore to guide users with virtual arrows overlaid on the live camera view.
- Implemented pointer animations to direct users from a starting point to a destination in real-time AR overlays, providing intuitive, step-by-step visual guidance.
- Delivered as Phase I of a larger navigation project, establishing the core AR guidance pipeline that later phases can build on.
- Integrated ARCore for motion tracking and environment understanding so that virtual arrows stay anchored to the real-world surroundings as the user moves.
- Made a mobile interface with a Unity-based AR view to keep the user experience simple while handling real-time 3D rendering.

Tic Tac Toe Game with Java and JDBC

- Created a Java-based Tic Tac Toe game with JDBC integration to store player data, scores, and game history in a relational database.
- Designed the database schema and used JDBC queries to save and retrieve match results, allowing players to review past games and cumulative scores.
- Implemented game logic in Java covering turn handling, win and draw detection, and input validation for a smooth two-player experience.
- Practiced core software engineering concepts including object-oriented design, exception handling, and database connectivity through a complete end-to-end application.

CERTIFICATIONS

- Introduction to Large Language Models — NPTEL Certification
- Deep Learning — NPTEL Certification
- Introduction to GitHub Copilot — Microsoft
- Java Programming 2022 — Infosys Springboard
- Introduction to IoT and Digital Transformation — Cisco Networking Academy
- Body Language – Key to Success — NPTEL Certification